

BRIGETTE VAZQUEZ SEGOVIA

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EDUCATION

University of Michigan

August 2024–Present

Ph.D., Astronomy and Astrophysics, expected 2029

Ann Arbor, MI

University of California, San Diego

September 2020–June 2024

B.S., Physics with Specialization in Astrophysics; Minor in Mathematics

San Diego, CA

RESEARCH EXPERIENCE

University of Michigan

August 2024–Present

Graduate Student Researcher *with Professor Monica Valluri*

- Develop modified particle-spray simulations to study how the time-evolving tidal field, mass, inner density profile, and orbit of a parent dwarf-galaxy dark matter halo shape the morphology and kinematics of globular-cluster stellar streams, including gaps, spurs, and tidal heating.

Harvard Center for Astrophysics

June 2023–September 2023

Astronomy Summer Intern *with Dr. Priyanka Chakraborty and Dr. Adam Foster*

- Analyzed Chandra LETG observations of the active galactic nucleus NGC 3227 using XSPEC and Cloudy spectral-synthesis models to constrain the photoionization parameter and column density.

Carnegie Observatories CASSI REU

June 2022–June 2024

with Dr. Ana Bonaca

- Used simulated Milky Way-like systems to investigate how cold dark matter subhalo encounters perturb stellar streams, reconstructing subhalo orbital histories and quantifying their effects on stream kinematics.

University of California, San Diego

June 2021–June 2022

with Professor Adam Burgasser

- Characterized Gaia eDR3 wide binary systems with late-M and L dwarf companions using multi-wavelength survey data and SpeX near-infrared spectral reductions.

PUBLICATIONS

Segovia, B., Valluri, M., & Chen, Y. “Did a parent dwarf galaxy produce the spur in the accreted globular cluster stream GD-1?” 2026, *The Astrophysical Journal*, submitted.

Zhou, T., Jacobsen, D., **Vazquez Segovia, B.**, et al. “Resolved Binaries with Late-M and L Dwarf Companions Identified in Gaia eDR3.” 2023, *Research Notes of the AAS*, 7, 50.

Segovia, B. V., Bonaca, A., & Nadler, E. “Detecting Dark Matter Subhalos Using Gaia-1 and Other Stellar Streams.” 2023, *Bulletin of the AAS*, 55, 2.

SCIENCE PRESENTATIONS

Carnegie Mellon University, Pittsburgh, PA

June 2024

Accreted Globular Cluster Stellar Stream Microstructures Resulting from Tides from the Parent Dwarf Galaxy

243rd American Astronomical Society Meeting, New Orleans, LA

January 2024

Constraining the Physical Conditions in Active Galactic Nuclei: A Case Study of NGC 3227 [Poster]

Harvard-SAO Astrophysics REU Symposium, Cambridge, MA

August 2023

Constraining the Physical Conditions in Active Galactic Nuclei: A Case Study of NGC 3227

241st American Astronomical Society Meeting, Seattle, WA Detecting Dark Matter Subhalos Using Gaia-1 and Other Stellar Streams [Poster]	<i>January 2023</i>
Carnegie Observatories REU Symposium, Pasadena, CA Detecting Dark Matter Subhalos Using Gaia-1 and Other Stellar Streams	<i>August 2022</i>
UC San Diego Lightning Talk, San Diego, CA Curating a Sample of M and L Dwarfs in Wide Binary Systems	<i>February 2022</i>

AWARDS & RECOGNITION

2024–Present **Rackham Merit Fellow**, University of Michigan
 2023 **Harvard Astronomy Summer Intern Program**, Harvard Center for Astrophysics
 2022 **Carnegie CASSI REU Internship**, Carnegie Observatories
 2021 **Triton Research Scholar**, University of California, San Diego
 2020–2024 **Provost Honors**, University of California, San Diego
 2020–2024 **Morgan Family Foundation Scholarship**

TEACHING

UC San Diego Supplemental Instruction Leader *September 2022–June 2024*

- Led weekly study sessions for undergraduate physics and programming courses.
- Facilitated collaborative learning, developed Python coursework, and supported students in strengthening problem-solving skills.
- Coordinated with faculty and conducted review sessions before exams.

OUTREACH & SERVICE

Undergraduate Research Advising *Fall 2024–Present*
 Graduate Student Advisor

- Support an undergraduate student through monthly advising meetings focused on research progress, coursework, internship applications, and professional development.

Stellar Halos Group Meetings *Fall 2025–Present*
 Meeting Host and Organizer

- Plan and host regular research group meetings for the stellar halos group, including coordinating meeting logistics, setting up the meeting space, and communicating with participants.
- Coordinate with invited guest speakers to schedule presentations and ensure smooth meeting organization.

Graduate Student Lunch Talks *Fall 2025–Present*
 Graduate Lunch Talk Host

- Serve as MC for weekly graduate student research lunch talks, introducing speakers and facilitating the presentation schedule.
- Coordinate with graduate students to organize the weekly speaker rotation.

University of Michigan Astronomy FEMMES *August 2024–Present*
 Coordinator

- Coordinate departmental outreach efforts for FEMMES, developing astronomy activities that introduce K–12 students to scientific inquiry, observational thinking, and careers in STEM.
- Design accessible presentations and hands-on demonstrations that connect astronomical concepts to students' everyday experiences.

Senior Outreach Center *February 2024–Present*

Student Volunteer

- Collaborate with astronomy graduate students to create community-centered astronomy programming for senior citizens in Ann Arbor.
- Develop presentations and interactive activities that make current topics in astronomy accessible to lifelong learners.

UC San Diego Astronomy Club

January 2023–June 2024

Vice President

- Helped reestablish the UC San Diego Astronomy Club as an active student organization by organizing stargazing nights, outreach events, and community-building activities.
- Led outreach planning, media coordination, and faculty communication to strengthen student engagement with astronomy.

UC San Diego Society of Physics Students

September 2020–June 2024

External Secretary and Outreach Member

- Supported programming and outreach efforts aimed at building an inclusive physics community for undergraduate students.
- Organized events, maintained communication calendars, and helped coordinate faculty talks and student-facing academic resources.

SKILLS

PROGRAMMING & SCIENTIFIC COMPUTING: Python, MATLAB, IDL

TECHNICAL SKILLS: Numerical simulation, data analysis, scientific visualization, and computational modeling

LANGUAGES: Native fluency in English and Spanish; beginner French